

Sample Exam

Medical Image Analysis

Student name:_____

Immatriculation number:_____

Exam Questions

Note: All the questions have equal weight.

1. A measurement taken in an image corresponds to a real-world quantity only through the voxel size. An anatomical structure is segmented in two scans of the same patient, acquired one year apart on different scanners.
 - (a) In the *baseline* scan the voxel size is $0.8 \times 0.8 \times 3.0 \text{ mm}^3$ and the segmentation contains 4 500 voxels. What is the volume of the structure, in mm^3 and in cm^3 ?
 - (b) In the *follow-up* scan the voxel size is $0.5 \times 0.5 \times 2.0 \text{ mm}^3$ and the segmentation contains 16 000 voxels. What is the volume now, and has the structure grown?
 - (c) What would you have concluded if you had compared the raw voxel counts instead, and what does this tell you about comparing measurements across acquisitions?
2. Deep learning libraries provide a layer called “convolution”, but the operation they implement is usually the closely related cross-correlation. For an image I and a kernel k , which of the two expressions below is the *convolution* $I * k$? Please circle the correct choice.

$$\sum_m \sum_n I(x - m, y - n) k(m, n)$$

$$\sum_m \sum_n I(x + m, y + n) k(m, n)$$

3. A great many methods in this course have the same shape: a data term measuring agreement with the observation, plus a regularisation term expressing which solutions are plausible,

$$\min_{\theta} D(\text{data}; \theta) + \lambda R(\theta).$$

For each of the three methods below, please state what plays the role of the data term and what plays the role of the regulariser.

(a) Total variation denoising.

(b) Deformable intensity-based registration with linear-elastic regularisation.

(c) MAP segmentation with a Potts / Ising Markov random field prior.

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.

(a) True False Convolution is translation equivariant but not translation invariant.

(b) True False A “same” (padded) convolution with a $k \times k$ kernel and stride 1 reduces each spatial dimension by $k - 1$.

(c) True False Fully connected layers are naturally translation invariant, which is why they are placed at the end of a classification network.

(d) True False For a recognition task two images differing only by a translation should give identical outputs, whereas for segmentation they should give the same output at a translated location.

5. A multi-head self-attention block has input dimension $D = 512$ and $H = 8$ heads, with per-head dimensions $D_K = D_Q = D_V = 64$. Counting the projection matrices $W_{K_h}, W_{Q_h}, W_{V_h}$ of every head together with the output projection W_0 , and ignoring all biases, how many parameters does the block have?
6. Once a Markov random field prior is added, segmentation can no longer be solved by deciding each pixel independently. Please explain briefly why exact inference becomes hard, and describe one approximate method presented in the lecture. You do not need to write down the equations.
7. Which of the following statements on the differences between Active Shape Models (ASM) and Active Appearance Models (AAM) is **inaccurate**?
 - ☐ (A) ASM searches *around* the current position, sampling along the surface normals of the model points, while AAM evaluates its fit *at* the current position.
 - ☐ (B) ASM minimises the distance between the model points and the corresponding points found in the image, whereas AAM minimises the difference between a synthesised model image and the target image.
 - ☐ (C) AAM usually has a larger capture range than ASM, because the additional texture information gives it more to lock onto when the initialisation is poor.
 - ☐ (D) AAM builds its texture model by warping every training image to the mean shape before sampling and normalising the intensities.
8. Which of the following statements on non-linear transformation models is **incorrect**?
 - ☐ (A) A naive parameterisation with one independent displacement vector per voxel can generate transformations that are physically implausible and even change the topology of the anatomy.

- ☐ (B) In a flow model each point follows the ordinary differential equation $dx/dt = v(x, t)$, and if the velocity field is smooth the resulting transformation is guaranteed to be a diffeomorphism.
 - ☐ (C) A stationary velocity field has more parameters than a time-dependent one, which is why the scaling-and-squaring scheme is needed to make it tractable.
 - ☐ (D) The diffusion model requires the displacement field to satisfy the Poisson equation, $\Delta u(x') + F(x') = 0$.
9. Which of the following statements on intensity normalisation is **incorrect**?
- ☐ (A) Min-max normalisation and matching the mean and standard deviation are both linear mappings of the intensities, and neither can change the contrast of an image.
 - ☐ (B) Nyul's method matches the *locations* of landmarks of the histograms, which aligns the distributions without making the two histograms identical.
 - ☐ (C) Histogram matching exploits the spatial correlation between neighbouring pixels, which is what makes it robust to differences in the size of the imaged object.
 - ☐ (D) Intensity variation is a particular problem for MRI because it has no absolute intensity scale, unlike CT where intensities are expressed in Hounsfield units.
10. In patch-based label fusion the label at a target voxel is decided by $L(x) = \arg \max_l \sum_i w_i \delta(\text{label}_i = l) / \sum_i w_i$, where the weights w_i come from the similarity between the patch around the target voxel and patches taken from the atlas library. Which of the following statements about this scheme is correct?
- ☐ (A) The weights are computed from the spatial distance between the target voxel and the atlas voxel, not from the image content.
 - ☐ (B) Only a coarse rigid or affine registration of the atlases is required, because correspondence is re-established locally by comparing patches inside a search window around each voxel.
 - ☐ (C) The scheme requires exactly one atlas; with several atlases the weights no longer sum to one and the decision becomes ill-defined.
 - ☐ (D) Because the weighting is borrowed from non-local means, the scheme can only be applied to denoising and not to the propagation of labels.
 - ☐ (E) The search window has to cover the whole image, since restricting it would bias the weights.
11. A U-Net for segmentation takes 128×128 inputs with one channel. All convolutions are 3×3 with “same” padding; each encoder level ends with 2×2 max pooling of stride 2, and each decoder level begins by upsampling by a factor of 2. The encoder levels produce 32, 64 and 128 channels respectively, and the bottleneck produces 256 channels. Each skip connection concatenates the encoder features onto the upsampled decoder features at the matching level. Circle True or False:
- (a) True False The feature maps at the bottleneck have a spatial size of 16×16 .
 - (b) True False Because all convolutions use “same” padding, no spatial information is lost in the encoder, and the skip connections are therefore redundant.
 - (c) True False After upsampling the bottleneck to 32×32 and concatenating the skip connection from the matching encoder level, the first decoder convolution at that level receives $256 + 128 = 384$ input channels.

- (d) True False A 3×3 convolution mapping those 384 input channels to 128 output channels has $3 \cdot 3 \cdot 384 = 3\,456$ weights, excluding biases.

12. Which of the following statements on interpretability and its uses are correct (multiple answers are possible)?

- ☐ (A) Saliency maps change over the course of training, and can be used as a kind of fingerprint of how a model responds to its inputs.
- ☐ (B) In interpretability-guided active learning, a sample is treated as informative when the model appears to struggle with it; easy samples tend to produce clean, focused saliency.
- ☐ (C) A model-agnostic method such as LIME explains a prediction by fitting a simpler, interpretable model that approximates the classifier in a neighbourhood of the input, weighted by a proximity measure.
- ☐ (D) Class-prototype maximisation returns the image from the training set that most strongly activates a given class.
- ☐ (E) Grad-CAM can be applied without any access to the internals of the model, since it only requires the class scores.